

AMLAN SARKAR

B.Tech CSE Student | Data Analytics | Python | SQL

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SUMMARY

Data Analytics-focused B.Tech CSE student with internship experience in Data Analytics and Machine Learning. Skilled in SQL, Python, Pandas, NumPy, data cleaning, exploratory data analysis (EDA), data visualization, feature engineering, statistical analysis, and predictive modeling. Experienced in analyzing datasets, deriving business insights, evaluating predictive models, and presenting findings through interactive dashboards with Streamlit.

EDUCATION

Sarala Birla University, Jharkhand

2023 – 2027 (Expected)

B.Tech in Computer Science & Engineering | CGPA: 7.65 / 10

EXPERIENCE

Data Analytics Intern, IBM SkillsBuild | AICTE

Aug 2026 – Sep 2026

- Analyzed 14,204 sales records across 1,559 products and 10 outlets to identify sales drivers and predict item-level outlet sales.
- Found that Outlet Type (41.3%), MRP Segment (30.4%) and Item MRP (23.4%) were the strongest sales drivers, translating model findings into actionable pricing and outlet-level insights.
- Delivered an interactive Streamlit application with prediction, model performance, and methodology views, supported by documented analysis and reproducible model artifacts.

Machine Learning Intern, IIIT Ranchi (IEEE Sponsored)

May 2026 – Jul 2026

- Developed an end-to-end stock market prediction system using historical stock data, return-based features, and a 60-day lookback window.
- Built and evaluated six forecasting approaches: BiLSTM, GRU, XGBoost, Random Forest, Linear Regression, and weighted Ensemble using RMSE, MAE, MAPE, R^2 , and directional accuracy.
- Achieved R^2 of 0.8510 with both BiLSTM and Random Forest and integrated SHAP-based model interpretability into an interactive Streamlit dashboard.

PROJECTS

BigMart Sales Intelligence & Prediction [\[Live Demo\]](#)

Python, scikit-learn, Streamlit

- Selected an Extra Trees Regressor using grouped 5-fold cross-validation, achieving $R^2 = 0.598$ and MAE = INR 757.
- Performed EDA and feature engineering to uncover sales patterns across outlet type, pricing, product attributes, and outlet characteristics.
- Compared Linear Regression, Random Forest, Extra Trees, XGBoost, and LightGBM using consistent grouped folds and preserved model artifacts for reproducible predictions.

Stock Market Prediction System [\[Live Demo\]](#)

Python, XGBoost, SHAP, Streamlit

- Rebuilt the prediction pipeline around stationary, return-based features after diagnosing MinMaxScaler extrapolation issues.
- Applied SHAP to interpret feature contributions and analyzed directional accuracy across forecasting models.
- Developed an interactive dashboard for model comparison and feature-driver analysis.

Parkinson's Disease Detection System

Python, scikit-learn

- Built an inference pipeline handling class imbalance and PCA-based dimensionality reduction, with soft-voting ensemble classification.
- Authored the full project report covering problem definition, methodology, results and conclusions.

SKILLS

Data Analysis: SQL, Python, Pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA), Data Visualization, Statistical Analysis, Microsoft Excel

Machine Learning: scikit-learn, XGBoost, SHAP, Feature Engineering, Predictive Modeling, Model Evaluation, Cross-Validation

Tools: Streamlit, Git, GitHub, VSCode, Jupyter Notebook

Soft Skills: Analytical Thinking, Problem Solving, Communication, Attention to Detail, Adaptability, Critical Thinking

CERTIFICATIONS

IBM SkillsBuild: Getting Started with Data, Data Fundamentals, Generative AI Essentials: Using LLMs to Work with Data

Cisco Networking Academy: Data Analytics Essentials, Introduction to Data Science, Find Insights with AI, Python Essentials 1, Introduction to Modern AI

INTERESTS

Gaming, Music, Football, Badminton